

Supervised Learning Report

CS7641: Machine Learning

Summer 2026

1 Assignment Weight

The assignment is worth 15% of the total points.

Read everything below carefully as this assignment has changed term-over-term.

2 Objective

You will build a reproducible supervised learning evaluation harness for a single multiclass classification problem using the Forest Cover Type dataset.

Your work will produce:

1. an exploratory data analysis (EDA) summary of the full dataset and the sampled dataset,
2. a justified sampling design of approximately 20,000 instances from the full dataset,
3. baseline and tuned models for four supervised learning algorithm families,
4. learning and model-complexity curves,
5. runtime/efficiency profiling, and
6. a cross-model comparison grounded in evidence.

Your goal is not merely to obtain strong metrics. Your goal is to explain model behavior using evidence from EDA, class imbalance, curves, runtime, bias/variance diagnostics, and leakage-safe evaluation.

Hypothesis-driven analysis requirement

After completing EDA and selecting your approximately 20,000-instance dataset, you must propose at least **one clear hypothesis** about expected algorithmic performance before running full tuning and final experiments.

Your hypothesis must be grounded in theory from the lectures and readings, such as bias–variance, margin-based classifiers, sensitivity to scaling, curse of dimensionality, model capacity, regularization, locality effects, and class imbalance.

Your experiments and discussion should be structured around testing this hypothesis, not merely reporting results.

3 Evaluation Overview

The detailed grading rubric used by the TAs contains specific all-or-nothing criteria, but the main categories below summarize how your report will be evaluated. This table is intended to help you understand the relative emphasis of the assignment without replacing the full rubric.

Category	Points	What this means
Compliance	16	Correct Covertypes dataset and approximately 20,000-instance sample, required Enterprise Overleaf and GT Enterprise GitHub access, reproducibility materials, readable figures and tables, outside sources, and AI Use Statement.
Study Design	14	Full-data and sample EDA, sampling justification, metric justification, hypothesis, validation protocol, and leakage controls.
Algorithm Analysis	56	Decision Trees, kNN, SVM, and Neural Networks, including tuning, learning curves, model-complexity curves, runtime, final metrics, and algorithm-specific interpretation.
Discussion	14	Cross-model synthesis, performance and efficiency trade-offs, class-level behavior, curve-based reasoning, limitations, and evidence-based improvements.
Total	100	
Extra Credit	+5	Neural-network activation-function study using SGD only.

The report should not read as a checklist of the table above. The strongest reports will integrate these pieces into a coherent technical paper that explains what was done, why it was done, what happened, and what the results mean. Every figure and table should be readable and should support a specific claim in the report.

4 Required Dataset for Summer 2026

The required dataset for this assignment is the **Forest Cover Type** dataset, also known as **Covertypes**. If this dataset is not used, you will receive a zero for the assignment.

Forest Cover Type / Covertypes

- **Task type:** Multiclass classification
- **Target:** `Cover_Type`, with classes 1 through 7
- **Source:** UCI Machine Learning Repository, Canvas, or direct import using Python tools such as `sklearn.datasets.fetch_covtype`
- **Dataset size:** The full dataset contains more than half a million instances. You must create and analyze a smaller working dataset of approximately **20,000 instances**.
- **Features:** Cartographic variables including continuous topographic measures and binary wilderness/soil-type indicators.
- **Notes:** The dataset is multiclass and imbalanced. Classes 1 and 2 are much more common than several minority classes.
- **Required evaluation metrics:** Accuracy, Macro-F1, and balanced accuracy. You must also include a confusion matrix and per-class precision, recall, and F1 discussion.

Required sampling design

Since the full dataset contains over half a million instances, you will design a smaller dataset of approximately **20,000 instances**. Your report must justify this choice.

Your sampling section must include:

1. the size of the full dataset and the size of your sampled dataset,
2. the class distribution of the full dataset,
3. the class distribution of your sampled dataset,
4. a justification for random sampling, stratified sampling, or another defensible sampling strategy,
5. a discussion of how your sampling choice affects minority classes, especially the rarest cover types, and

6. the exact random seed and procedure used to create the sample.

A stratified sample is the default expectation since the target classes are heavily imbalanced. A simple random sample may be acceptable only if you demonstrate that it preserves all classes and approximately reflects the full class distribution. Any non-stratified sampling choice must be justified using evidence from EDA.

Do **not** use the first 20,000 rows as your working dataset unless you provide a compelling justification. Your sampling procedure must avoid accidental ordering effects.

EDA scope requirement

Detailed EDA tables and supporting plots may be placed in the reproducibility sheet. However, the main report must still stand alone. It must include the important EDA details needed to justify your sampling strategy, metric choices, hypothesis, preprocessing decisions, and interpretation of model behavior.

Important sanity requirement

If your metrics are *too perfect* or implausibly high, you must include a **Leakage / Evaluation Debug** subsection demonstrating that your evaluation is correct.

At minimum, include one diagnostic such as:

1. a shuffled-label baseline, where performance should collapse,
2. a dummy baseline, such as most-frequent class prediction, or
3. a train/test swap or split-integrity sanity check.

5 Procedure (What you will do)

You will complete **one supervised learning task**: multiclass classification of forest cover type.

You will:

1. load the full Covertypes dataset,
2. conduct EDA on the full dataset,
3. design an approximately 20,000-instance working dataset and justify the sampling strategy,
4. conduct EDA on the sampled dataset,
5. develop at least one hypothesis about expected algorithmic performance,
6. build a reproducible preprocessing and evaluation workflow,
7. train and tune the required algorithms, and
8. write a thorough analysis of your findings.

You do not need to implement any learning algorithm yourself. You may use standard libraries and code-assistance tools, but you are responsible for understanding, verifying, and explaining all code, experiments, figures, and analysis submitted.

Hypothesis-first workflow

Your hypothesis should make a concrete prediction about model behavior and should be informed by the sampled dataset's feature structure, class imbalance, scale sensitivity, and expected bias-variance behavior.

What counts as a well-posed hypothesis

A strong hypothesis includes:

1. a predicted ordering or trade-off, such as $\text{RBF SVM} \geq \text{kNN} \geq \text{DT}$,
2. a reason tied to dataset properties observed in EDA, such as feature scale, class imbalance, nonlinear structure, sparsity, or noise, and
3. a theory-based explanation, such as margin maximization, regularization, bias-variance, locality effects, or sensitivity to high dimensionality.

Avoid generic claims that cannot be tested.

General rules

- You may program in any language and are allowed to use standard libraries, **as long as they were not written specifically to solve this assignment**. Code repositories, notebooks, or AutoML templates created for CS7641 assignments (“plug-and-play” solutions) are **not allowed**.
- TAs must be able to recreate your experiments on a standard Linux machine if necessary.
- The analysis you provide in the report is paramount and the focus of the assignment.

6 Required Algorithms

You must evaluate the following learning algorithms on your sampled Covertypes dataset:

- **Decision Trees (DT)**. Use some form of pruning or regularization, such as post-pruning via *ccp- α* , depth limits, or leaf constraints. You are not required to use information gain. Gini or entropy are both acceptable. **State what you use and why**.
- **k-Nearest Neighbors (kNN)**. Compare **meaningfully different** values of k , such as small, medium, and large neighborhoods. Scale features where appropriate and justify your distance metric and weighting choice.
- **Support Vector Machines (SVM)**. Evaluate ≥ 2 **kernels**, such as linear and RBF. Tune C and γ where relevant. Since SVMs can be computationally expensive, you must report runtime and may use a clearly justified validation strategy.
- **Neural Networks (MLP; SGD only)**. You must train **two** neural network implementations:
 - **One scikit-learn model**: `MLPClassifier` using **SGD only**. For scikit-learn, use `solver='sgd'`, set `momentum=0`, and do not use Nesterov momentum.
 - **One PyTorch model**: a compact MLP trained with **SGD only**. Use plain SGD without momentum.

You will compare capacity scaling, such as depth versus width, while keeping SGD constant. When comparing architectures, keep the total parameter count approximately constant when feasible. **No momentum, no Nesterov, and no adaptive variants such as Adam, Adagrad, or RMSprop are allowed**. Report learning rate, batch size, epochs or early stopping, and regularization choices.

Minimum hyperparameter exploration standard

When comparing hyperparameters, you should evaluate at least **three distinct and reasonable values** for the relevant scalar hyperparameter unless you provide a clear computational justification for a smaller but still meaningful sweep.

A reasonable sweep should cover a range broad enough to reveal underfitting, overfitting, or performance saturation. Testing tiny variations around one value, such as $k = 5, 6, 7$, does not count as meaningful exploration unless justified by prior evidence.

For computationally expensive models, especially SVMs, a smaller sweep may be acceptable if you explain the computational constraint, report runtime evidence, and still compare enough settings to support your final model choice.

7 Required Workflow Outputs

Applies to the sampled Covertypes dataset and each required algorithm component: DT, kNN, SVM, NN-Sklearn, and NN-PyTorch.

Data, targets, sampling, and leakage controls

- Declare the target, task type, feature groups, and chosen metrics.
- Report the class distribution of both the full dataset and your sampled dataset.
- Justify your sampling design and discuss its implications for minority classes.
- State leakage controls you applied, including what you removed, what you verified, and why.
- Use a single held-out test split once at the end. Perform tuning using cross-validation or a train/validation split on the training data only.
- State at least one hypothesis before presenting tuned results, and explicitly revisit it in your conclusion.

Methodology & reproducibility

- Fix random seeds for sampling, splitting, model training, and neural network initialization where applicable.
- Record exact splits and make preprocessing deterministic.
- Use leakage-safe preprocessing, such as fitting scalers only on training data.
- Provide sufficient instructions to reproduce your results on a standard Linux machine.

Required figures/tables

- **EDA summary:** include class distribution, feature scale/range observations, and at least one dataset-specific observation relevant to model behavior.
- **Learning curves (LC):** training and validation metric vs. training size using at least **four** training sizes unless a clear computational justification is provided. Diagnose bias/variance, underfitting, overfitting, and whether more data appears likely to help.
- **Model-complexity curves (MC):** validation metric vs. one important hyperparameter, such as NN width/L2, SVM C/γ , kNN k , or DT depth/ ccp_alpha . Use at least **three** reasonable values for the selected hyperparameter unless you provide a clear computational justification. The curve should support your final model choice.
- **Runtime table:** fit and predict wall-clock times with a brief hardware note.
- **Multiclass evaluation:** confusion matrix at a justified operating point and per-class precision, recall, and F1 discussion.

Metric expectations

Since Covertypes is a multiclass and imbalanced classification problem, accuracy alone is insufficient.

At minimum, your report must include:

1. **Accuracy**, to summarize overall correctness,
2. **Macro-F1**, to evaluate performance across classes without allowing majority classes to dominate,
3. **Balanced accuracy**, to account for class imbalance in recall,
4. a **confusion matrix**, to identify which cover types are confused with one another, and
5. **per-class precision, recall, and F1**, with discussion of minority-class behavior.

Your metric discussion should explain what each metric reveals and why it is appropriate for this dataset. If a model has high accuracy but poor minority-class performance, your analysis must identify and discuss this limitation.

Algorithm-specific essentials

- **NN (both implementations)**: show epoch-based learning curves, such as train vs. validation loss or metric, and discuss early stopping and regularization.
- **NN comparison requirement**: explicitly compare scikit-learn vs. PyTorch results and discuss differences in optimization behavior, stability, runtime, and tunability under **SGD-only constraints**.
- **SVM**: scale features; use ≥ 2 kernels; tune C and kernel parameters where relevant.
- **kNN**: scale features; justify distance metric and weighting; discuss sensitivity to irrelevant features, binary indicators, and high dimensionality.
- **DT**: show a pruning or regularization curve and report final depth and number of leaves.

Cross-model discussion (required)

After presenting the individual algorithm results, you must include a discussion section that synthesizes what the full set of experiments reveals. Do not only present model-by-model results.

Your discussion should directly compare the algorithm families and explain why their behavior differed using evidence from final metrics, learning curves, model-complexity curves, runtime, and/or error patterns. Strong discussions should connect observed behavior to course concepts such as bias-variance, locality, margin-based classification, scaling sensitivity, model capacity, and regularization.

Helpful prompting questions:

- Which model family performed best overall, and does that conclusion change when you compare Accuracy, Macro-F1, and balanced accuracy?
- Did the best-performing model also have the best runtime, stability, or interpretability?
- Which algorithms appeared most sensitive to scaling, hyperparameters, or training size?
- Did the results match your original hypothesis? If not, what evidence explains the mismatch?
- What did the full set of experiments reveal that no single metric or model could show alone?

Trade-offs and class-level behavior (required)

Your discussion must explain trade-offs between predictive performance and computational cost. Include comparisons such as Accuracy/Macro-F1 versus fit time, prediction time for kNN, SVM kernel cost, tree interpretability versus performance, or neural-network training stability versus runtime.

Your discussion must also analyze multiclass performance rather than only aggregate scores. Identify which cover types were easiest or hardest to classify, discuss minority-class behavior, and use the confusion matrix or per-class precision/recall/F1 to explain at least one important misclassification pattern.

Helpful prompting questions:

- Which classes were most often confused, and what might explain those confusions?
- Did high overall accuracy hide poor performance on one or more minority classes?
- Did any model improve Macro-F1 or balanced accuracy while sacrificing overall accuracy?
- Was a slower model worth the runtime cost based on minority-class performance or error patterns?
- Did any model appear attractive numerically but weak once class-level behavior was examined?

Curve-based interpretation, limitations, and improvements (required)

Your discussion must use learning curves and model-complexity curves to make claims about underfitting, overfitting, variance, bias, saturation with more data, or hyperparameter sensitivity. These claims should be tied directly to the plotted curves.

You must also identify concrete limitations of the study and evidence-based improvements or follow-up experiments. Acceptable limitations include the 20,000-instance sample, minority-class instability, limited SVM grid size due to runtime, SGD-only neural-network constraints, class imbalance, or restricted feature engineering. The improvement must follow from the evidence, such as testing class weighting after poor minority recall, expanding the SVM grid after sensitivity to C , increasing minority-class representation, or revisiting NN learning rates after unstable epoch curves.

Avoid generic statements such as “use more data,” “try more models,” or “tune better” unless you connect the improvement to evidence from your results.

Helpful prompting questions:

- Did any learning curve suggest that more training data would likely improve validation performance?
- Did any model show high variance, high bias, or signs of saturation?
- Which hyperparameters had the clearest effect on performance, and which appeared relatively insensitive?
- What limitation most affects your confidence in the final comparison?
- What follow-up experiment would you run next, and what result from your report motivates it?

Conclusion (required)

Your conclusion should provide a concise synthesis of what was learned. It should not introduce new experiments or repeat every number from the results section.

Discuss:

- Type I/II-style errors in the context of multiclass classification, including which cover types are most often confused,
- what you learned about model behavior on the Covertypes dataset,
- whether your hypothesis was supported, contradicted, or only partially supported,
- how class imbalance affected interpretation, and
- one realistic next step.

Helpful prompting questions:

- What is the most defensible final model choice, and what evidence supports it?
- What is the most important caveat to that choice?
- What did this experiment teach you about supervised learning beyond this specific dataset?
- What should a future researcher or practitioner be careful about if using these results?

8 Extra Credit (up to 5 points; NN only; *SGD only*)

- **Activation function study.** Using the sampled Covertypes dataset, evaluate **four** activation functions, such as ReLU, GELU, SiLU/Swish, and tanh, using the *same* architecture, preprocessing, regularization, and training protocol. Use **SGD only**. *Deliverables:* epoch-wise validation curves for each activation, a comparison table of final metrics, and a brief analysis of convergence speed, stability, and trade-offs under SGD.

9 External Sources & Literature (required)

- Include outside sources beyond course materials, with at least **two peer-reviewed references**.
- Use these sources to **justify choices**, such as metrics for imbalance, sampling design, pruning, kernel decisions, leakage controls, and interpretation of results.
- Use **IEEE citation style**. Your references should be formatted consistently with the IEEE conference template.

10 Report Requirements

Your report is a formal technical paper limited to **8 pages total**, including figures and references. Anything past 8 pages will not be read.

You must use the official IEEE conference paper template for the report. In LaTeX, this means using the IEEE conference format, such as:

```
\documentclass[conference]{IEEEtran}
```

or the equivalent official IEEE conference template provided through Overleaf.

A generic **article** template, non-IEEE report template, or custom format is not acceptable for the report. The IEEE conference template requirement applies to **SL_Report_{GTusername}.pdf**. The reproducibility sheet does not need to use the IEEE conference template, but it must be clear, complete, and readable.

Your report must be created using your **Georgia Tech / Enterprise Overleaf account**. This is required to preserve versioning and provenance. When submitting your report, you are required to include a **READ-ONLY** link to the Overleaf project. If a link is not provided in the report or Canvas submission comment, the report will receive a 0 until the Overleaf project can be verified. Do not share the project directly via email.

Main report vs. reproducibility sheet

Detailed EDA tables, extended plots, environment details, and additional reproducibility information may be placed in the reproducibility sheet. However, the main report must still stand alone as a formal technical paper.

The main report must include enough EDA evidence to justify:

1. the sampling strategy,
2. the chosen metrics,
3. the hypothesis,
4. the preprocessing workflow, and
5. the interpretation of model results.

Do not move all substantive EDA interpretation to the reproducibility sheet. The reproducibility sheet supports the report, it does not replace the report.

Interpret, don't summarize

Every figure/table must include a takeaway tied to algorithm behavior and the data. Avoid repeating the legend, explain what the result *means* and why it happened.

Formal writing requirement

Your *report* must read like a technical paper, not a checklist.

Bullets are allowed when they improve clarity, such as listing hyperparameters, summarizing preprocessing steps, or describing an experimental protocol. However, your **analysis and conclusions must be written in paragraphs** with coherent reasoning.

Excessive bullet-only writing that replaces interpretation, especially in Results or Discussion sections, will be treated as a draft rather than a complete technical report. As a rule of thumb, multiple instances of four or more consecutive bullets in analysis-heavy sections will result in the report not being scored and being treated as a draft. If a section could be pasted into a spreadsheet and still make sense, it is not sufficient prose.

11 Acceptable Libraries

You may use any standard open-source libraries that support reproducible experimentation and were not written specifically to solve this assignment. The list below includes common examples.

Recommended stack (examples)

- **Core ML:** `scikit-learn` (pipelines, cross-validation, metrics, calibration), `imbalanced-learn` (imbalance utilities), and `PyTorch` (required).
- **Data & plotting:** `pandas`, `polars`, `NumPy`, and `matplotlib` (or `plotly/altair/seaborn`).

Neural network optimizer rule (required)

For **all neural network experiments**, you must use **SGD only**. Momentum, Nesterov acceleration, and adaptive optimizers (Adam/Adagrad/RMSprop/etc.) are not allowed.

For `scikit-learn` `MLPClassifier`, this means `solver='sgd'`, `momentum=0`, and `nesterovs_momentum=False`. For `PyTorch`, this means using plain SGD without momentum.

12 Submission Details

All scored assignments are due by the time and date indicated on Canvas (Eastern Time, ET). Canvas displays times in your local time zone if your profile is set correctly; grading deadlines are enforced in ET.

All assignments are due at 11:59:00 PM ET on the final Sunday of the unit. Submissions received by **7:59:00 AM ET** the next morning are accepted **without penalty**.

Late penalty

After the grace window, the score is reduced **20 points per calendar day** starting at 7:59:00AM ET day-over-day. Treat **11:59 PM ET** as your real deadline, allow time for upload and verification.

What to submit

You will submit **two PDFs**:

1. **SL_Report_{GTusername}.pdf** — your formal report, written using the **IEEE conference template** in your Georgia Tech / Enterprise Overleaf account.
2. **REPRO_SL_{GTusername}.pdf** — your reproducibility sheet, which must include:
 - (a) A **READ-ONLY** link to your Overleaf project.
 - (b) A **GitHub commit hash** (single SHA) from the final push of your code.

- (c) **Exact run instructions** to reproduce results on a standard Linux machine (environment setup, commands, data paths, and random seeds).
 - (d) The full-data and sampled-data EDA summaries.
 - (e) The exact sampling procedure used to create the approximately 20,000-instance dataset.
- Include the READ-ONLY Overleaf link in the report or Canvas submission comment. **Do not send email invitations.**
 - Use the **GT Enterprise GitHub and Georgia Tech / Enterprise Overleaf** for course-related code and report writing. Personal GitHub or personal Overleaf accounts are not acceptable for this assignment.
 - If you submit to Canvas multiple times, you might see an appended “-1” to the assignment name. This is acceptable because this is how Canvas is designed to handle multiple instances of the same file name.

13 Feedback Requests

When your assignment is scored, you will receive feedback explaining errors and successes. If you are confused by feedback, start a private thread on Ed.

14 Plagiarism and Proper Citation

The easiest way to fail this class is to plagiarize. **Using the analysis, code, or graphs of others in this class is considered plagiarism.** We care about your analysis, it must be original and grounded in your own experiments.

If you copy any amount of text from other students, websites, or any other source without proper attribution, that is plagiarism. Citing is required but does not permit copying large blocks of text. All citations must use **IEEE citation style** for this assignment.

LLMs (disclosure required)

We treat AI-based assistance the same way we treat collaboration with people. You may discuss ideas and seek help from classmates, colleagues, and AI tools, but all submitted work must be your own. For this course, a large language model is any model with more than one billion parameters. These tools can increase productivity, but they are aids, not substitutes for your skills. The goal of reports is synthesis of analysis, not merely getting an algorithm to run. Even if you locate code elsewhere or generate code with an AI tool, you must apply it thoughtfully to your data and write the analysis yourself.

Every submission must include an AI Use Statement. List the tools used and what they assisted with and confirm that you reviewed and understood all assisted content. If an editor or platform surfaces suggestions, you may keep them only with disclosure and verification. You are not required to disable assistants. Be aware that coding editors often include assistants by default. Google Colab and Visual Studio Code commonly surface them. Also check any writing or grammar platforms you use. Overleaf through Georgia Tech is acceptable.

Allowed with disclosure. Brainstorming, outlining, grammar and clarity edits, code generation, code refactoring, and debugging.

Not allowed. Submitting AI-written analysis, conclusions, or figures as your own. Fabricating results or citations. Paraphrasing AI or prior work to evade checks. Uploading course materials or private data to public tools.

Example Statement at the very end of the report before References. “AI Use Statement. I used ChatGPT and Visual Studio Code Copilot to brainstorm and outline sections of the report, generate and refactor small code snippets, debug a pandas indexing issue, and edit grammar and clarity throughout. I reviewed, verified, and understand all assisted content.”

Verification will rely on provenance and reproducibility. We will not ask for live coding. We will review platform histories, including Overleaf revision timelines and copy-and-paste patterns, and Git commit and branch history. We may request your repository with full history, Overleaf project history, notebooks, logs, intermediate outputs, and scripts used to generate figures and tables. Similarity and AI indicators, including Turnitin, are corroborating signals only and are not used in isolation.

If we suspect a violation, we will document evidence, notify you, and request artifacts. Cases may be referred to the Office of Student Integrity (OSI). Undisclosed or prohibited use is academic misconduct. This policy may be updated during the term. When in doubt, always use proper citations, and ask clarifying questions on Ed.

15 Version Control

- v1.0 – 05/19/2026 – TJJ revised SL Report for Summer 2026 term using the Covertypes multiclass classification task, student-facing evaluation overview, and IEEE conference template requirement.

Assignment description written by Theodore LaGrow.